Charles Perrault

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Summary

Aerospace-focused engineering student at Purdue University with hands-on experience in AI research, high-altitude balloon flight operations, and competitive robotics. Skilled in CAD, wiring, and programming, with field experience in launch operations and payload recovery.

Education

Purdue University — West Lafayette, IN
First-Year Engineering (Aerospace Engineering intended) — *Class of 2030*

Bioscience High School — Phoenix, AZ
Relevant Coursework: Engineering Design, Physics, Biology, Chemistry, Spanish 1-6 — *Graduated May 2026*

Experience

AI Research Assistant — 2025
YDSL, in collaboration with Stanford University faculty — *Phoenix, AZ*
- Conducted applied artificial intelligence research at YDSL under the mentorship of Stanford University professors
- Collaborated with a research team to investigate and evaluate AI methods and approaches
- Contributed to research workflows spanning data preparation, experimentation, and analysis

High-Altitude Weather Balloon Program — 2026
Launch Operations & Payload Retrieval — *Arizona*
- Served on launch operations team for a high-altitude weather balloon mission, executing pre-launch checks and coordinating a safe release
- Led post-flight payload retrieval using tracking data, gaining direct field experience with flight instrumentation and telemetry

Intern — 2025
College Depot — *Phoenix, AZ*
- Completed an internship supporting day-to-day operations, developing professional workplace, communication, and problem-solving skills

Robotics & Technical Leadership

Social Media Coordinator — 2025 – 2026
FIRST Robotics Competition – Team Epsilon 11100 — *Phoenix, AZ*
- Managed team's social media presence and communications during the 2026 competition season
- Team earned **Rookie All-Star Award** (Utah) and was recognized as **Best Rookie Team** (Arizona)

Robotics Team Member — 2023 – 2025
FIRST Robotics Competition – Team 2375 — *Phoenix, AZ*
- Programmed and wired the competition robot for the 2024/2025 seasons
- Competed at the 2024 FRC World Championships in Houston, TX

Social Media Coordinator — 2025 – 2026
Venture Crew, Troop 2041 — *Phoenix, AZ*
- Managed social media presence and helped organize crew outings and fundraisers
- Contributed 2 years of community service, developing leadership and event-planning skills

Awards & Certifications

Makers of Change: 1st Place, 2024–2025 Season; Competitor, 2025–2026 Season
Standardized Testing: 1570 SAT | 36 ACT | National Merit Semifinalist: 1520/1520 PSAT
Certifications: Wilderness First Aid (WFA) | Autodesk Fusion 360 Certified User | Arizona CTE Certificate of Technical Skill Attainment (2025)

Technical Skills

Languages: Java, Python
Engineering & CAD: Autodesk Fusion 360, Engineering Design
Tools & Software: LaTeX, Microsoft Office, Google Workspace, Web Development
Field Skills: Launch Operations, Payload Tracking & Retrieval, Wilderness First Aid
Core Strengths: Technical Leadership, Cross-Team Collaboration, Communication, Problem Solving